

HyMeKo Visual Atlas

Figures, plots, tables, and evidence map from the accessible workspace

Codex visual companion for Dr. Csaba Hajdu

2026-06-30

Abstract

This is the visual companion to `reports/2026-06-30-hymeko-exploration-summary.pdf`. It explains the current HyMeKo state mostly through figures, plots, compact tables, and evidence captions. It only uses artifacts accessible in this workspace: reports, figures, benchmark CSVs, JSON summaries, and the current verification run.

1 One-page evidence map

This page is the reader’s orientation key. It separates mature, publishable evidence from open research bets, and it names the figure family that supports each claim. The rest of the atlas expands each row into its figures, tables, and missing discriminator.

Area	Main visual evidence	What it says	Readiness
Cross-view IR	Cross-view grid and verification table	One HyMeKo source can emit multiple concrete views whose recovered invariants agree.	strong
Performance substrate	Scaling and control-loop tables	Core emission, expansion, and inference paths are fast enough to support tooling and demos.	strong
Topology matching	Topology map, zoo, arity plots	More edges do not automatically help; controller topology/arity must match the plant.	strong
RL/control	Overnight RL, toy worlds, BC/off-policy plots	BC works; PPO or off-policy refinement can collapse task success; fast toy lab is needed.	open
Vision/Soma	Readout ladder, spatial tree, RicciStim transfer	Position-aware readout partially rehabilitates Soma vision; readout matters more than holonomy.	mixed
FSR-LM	Architecture diagrams, bpb curve, speed curve	FSR pre-norm beats a tiny transformer on bpb; top-k spike gate fixes dense long-context wall.	open
Holonomy	Probe table	Pure cycle holonomy is multiplicative transport, not plain additive signed-walk aggregation.	open

Table 1: Compact map of the workspace’s current evidence.

2 Cross-view consistency: the publishable kernel

This is the strongest near-term contribution because it is a property of the working system, not a training run. The verifier treats each emitted view as a separate concrete syntax, extracts the same invariant, and checks agreement against the IR-level query. That makes model drift a falsifiable engineering failure rather than a vague documentation risk.

3 Core scaling and control-loop substrate

The infrastructure layer matters because the research claims sit on top of a real compiler, transform registry, query engine, and tensor substrate. These tables are not the paper’s main novelty, but they show that the framework can run the workflows that the papers and demos require.

HyMeKo cross-view consistency — machine-verified against the real emitters

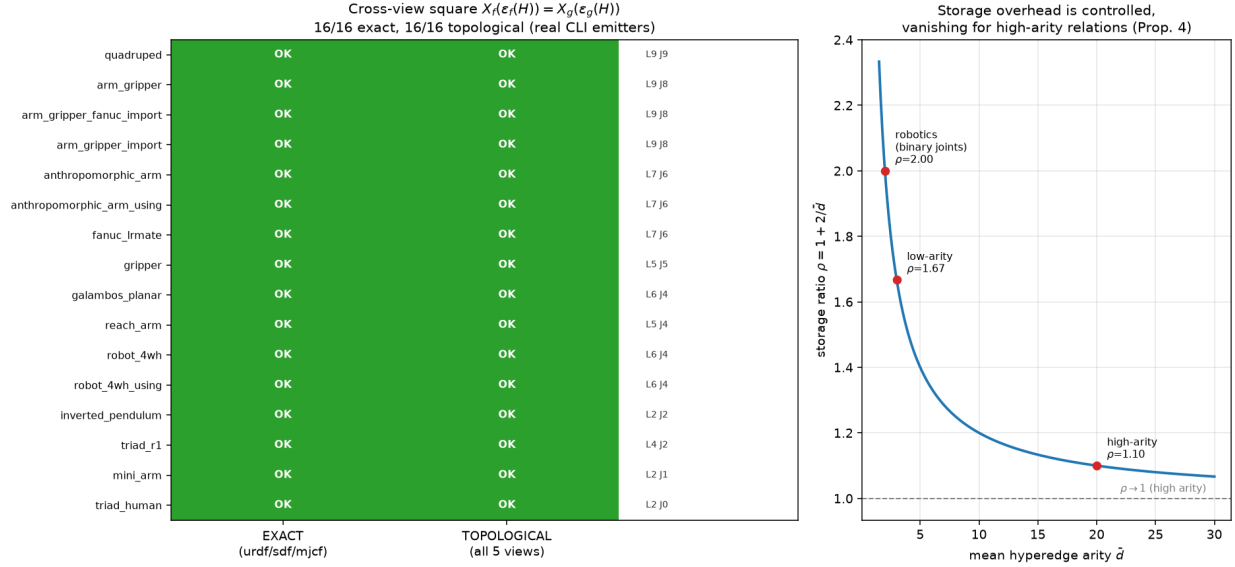


Figure 1: Machine-verified cross-view consistency. The key claim is a commuting square over real emitted views: parse the concrete view back to an invariant, and it agrees with the IR-level invariant.

Source: reports/figures/cross_view_consistency.png;reports/2026-06-28-cross-view-consistency.md

Domain	Views checked	Recovered invariant
Robotics / kinematics	URDF, SDF, MJCF, DOT, Mermaid	Links, masses, actuated joints, parent/child polarity, joint axes; exact for interchange views, topological for diagram views.
Systems engineering	SysML requirements, DOT requirements	Requirements, blocks, satisfy/allocate/derive relations, coverage, DAG/forest-style global properties.
Drift demo	Single-source HyMeKo vs pairwise manual files	HyMeKo re-emission keeps views synchronized; pairwise editing drifts by construction.

Table 2: The central formal-methods story is not a benchmark score; it is machine-checkable consistency across concrete views.

Artifact type under reports/	Count	Meaning
Markdown reports	345	Acceptance records and experiment narratives.
JSONL logs	78	Per-seed or per-run raw experiment traces.
PDF reports	61	Built technical reports, plans, and paper-style notes.
LaTeX sources	56	Rebuildable source for technical notes.
JSON summaries	15	Compact numeric outputs and benchmark summaries.
PNG figures at report root	13	Directly embedded summary plots.

Table 3: The repo is not just code; it is an evidence archive. This matters because most claims can be traced to a report, figure, and raw-ish artifact.

Source: PowerShellinventoryoverreports/

June report date	15	16	17	18	19	20	21	29
Markdown reports	17	7	10	12	14	14	13	12

Table 4: June 2026 had several dense experiment days. The atlas focuses on the highest-signal slices, not every report.

Source: PowerShellgroupingofreports/2026-06-*.md

Corpus	Compile median ms	Star expansion median ms	MP forward median ms
10 hyperedges	0.514	0.000201	0.00684
100 hyperedges	3.074	0.00129	0.0580
1000 hyperedges	30.852	0.0128	0.554
10000 hyperedges	465.336	0.169	13.072

Table 5: Selected scaling rows from the benchmark CSV. Expansion is tiny; message passing stays practical into 10k-edge synthetic fixtures.

Source: `hymeko_bench/results/scaling.csv`

Pipeline	Stage	Median us	P95 us
Hypergraph	state assembly	0.190	0.201
Hypergraph	forward	1.313	1.343
Hypergraph	total	1.593	1.624
TD3	total	0.431	0.441
PID+TD3	total	0.371	0.401

Table 6: Older control-cycle benchmark: the hypergraph path has overhead versus TD3/PID+TD3 but remains microsecond-scale in this harness.

Source: `hymeko_bench/results/control_cycle.csv`

3.1 Evidence inventory

4 Hypergraph views and tensor intuition

HyMeKo is most legible when the same object is viewed through several lenses: star expansion, clique expansion, tensor/incidence views, and generated target formats. These figures explain why the framework can talk simultaneously about formal views, graph learning, and control. They also explain why arity and view choice are not cosmetic: they change what information is preserved.

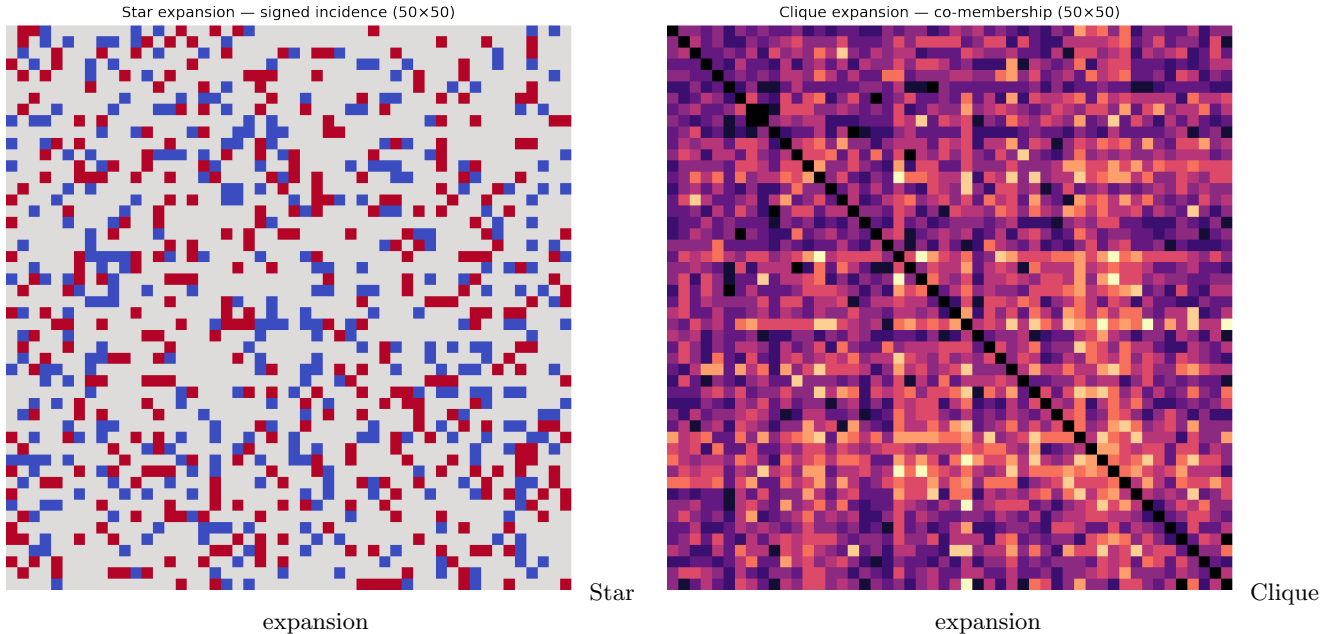


Figure 2: Two canonical graph views of a hypergraph. Star expansion preserves hyperedge identity via hub nodes; clique expansion collapses each hyperedge into pairwise relations. This visual distinction underlies much of the arity and view-consistency story.

Source: `docs/seminar/figures/star_expansion.png`; `docs/seminar/figures/clique_expansion.png`

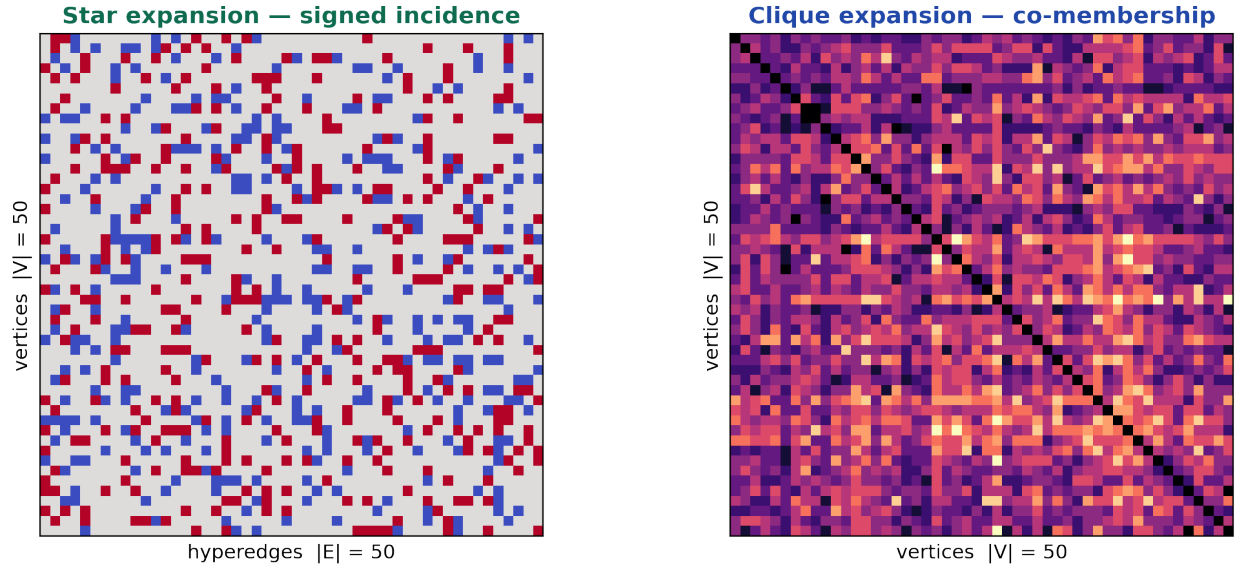


Figure 3: Tensor-view panels used in the seminar materials. These are the bridge from declarative signed hypergraphs to tensorized learning/control substrates: incidence, expansion, and message-passing views become concrete arrays.

Source: docs/seminar/figures/tensor_panels.png

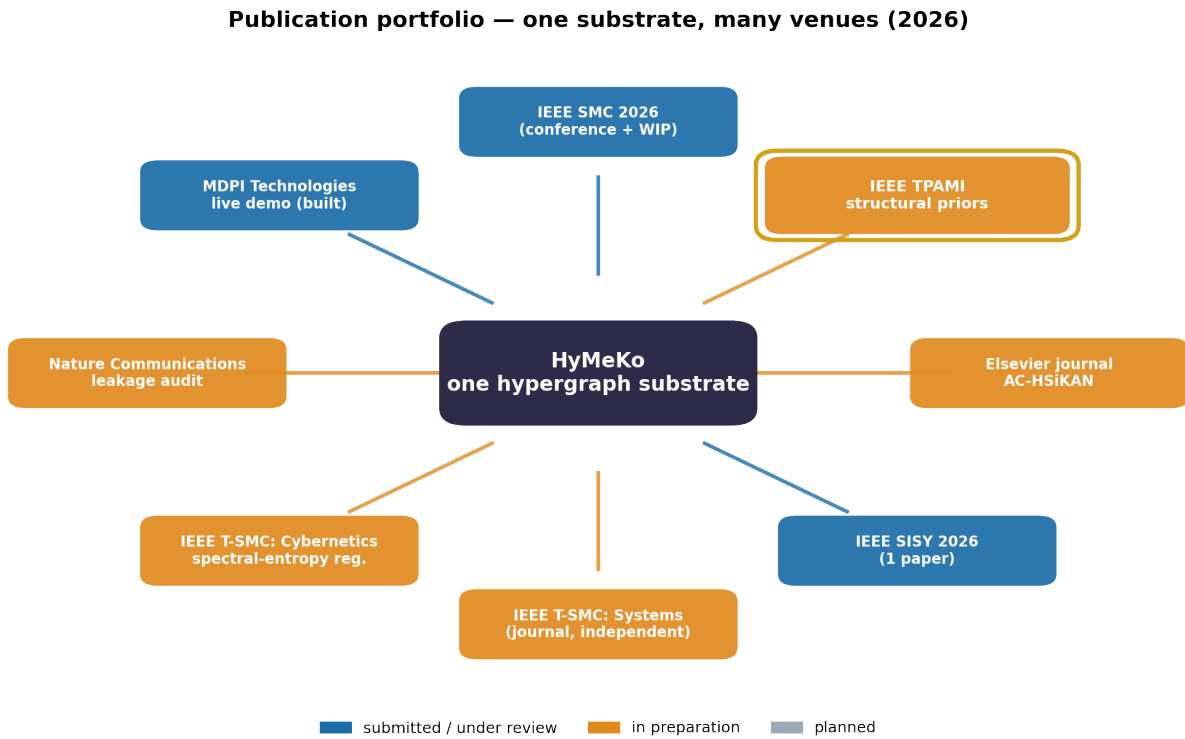


Figure 4: Publication/portfolio map from the seminar deck. It is useful as a visual reminder that the project has multiple venue families, not one monolithic paper.

Source: docs/seminar/figures/publications.png

5 Topology: the matching law

The topology line asks a control-facing question: if the plant has a coupling structure, should the controller mirror it? The supervised topology map says yes at proxy scale: the matched controller wins every row, while the densest complete controller does not act as a universal substitute. The important claim is not “this topology is best”; it is “match beats capacity.”



Figure 5: Topology-to-performance matrix. The matched controller wins every plant in this supervised structural proxy; the dense complete controller is not a universal solution.

Source: `reports/topology_map/topology_map.png`; `reports/2026-06-27-topology-performance-map.md`

6 Arity: hypergraph lift is not free

This section kills an attractive but wrong shortcut: lifting a graph plant into a hypergraph does not automatically improve compute or accuracy. The lift helps only when the plant itself is hypergraph-structured. That gives the project a cleaner law: match arity to the plant, instead of treating hypergraphs as a free upgrade over graphs.

7 RL/control: useful substrate, unresolved architecture win

The RL part is an experimental engine rather than a finished victory claim. The repo already has MuJoCo tasks, BC warm starts, PPO/off-policy refiners, GIF rendering, checkpoints, and fast physics-free sanity worlds. The key lesson is that task success and scalar return must be separated: a refinement run can make the reward curve look better while the policy stops completing the task.

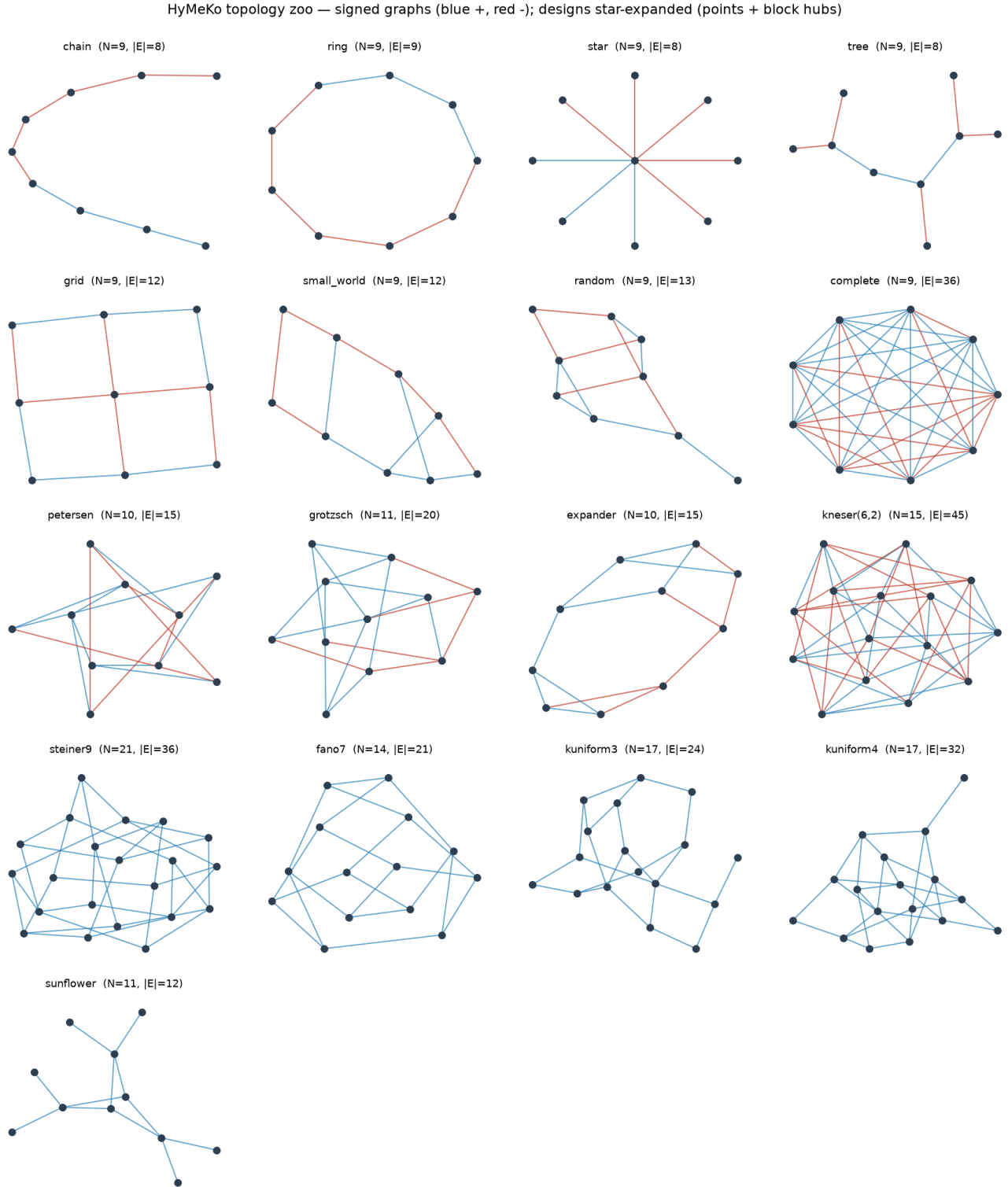


Figure 6: Topology zoo: generic graphs, combinatorial designs, and extremal high-symmetry constructions. These are candidate walk bases for structural controllers.

Source: [reports/figures/topology_zoo/gallery.png](#); [reports/2026-06-28-topology-zoo.tex](#)

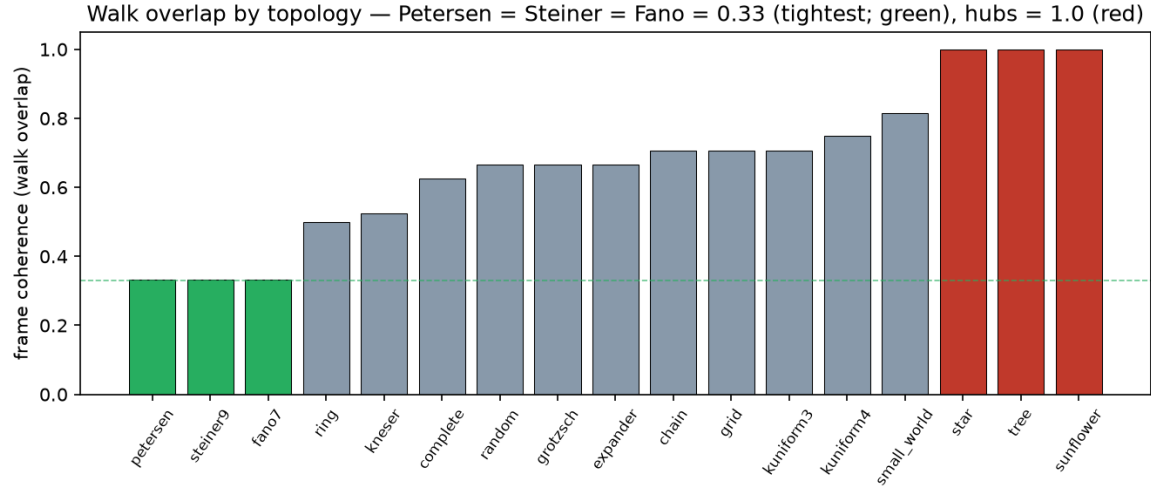


Figure 7: Walk-basis frame coherence across topologies. Low coherence looks mathematically attractive, but the companion control map shows that matching the plant dominates any universal topology ranking.

Source: reports/figures/walk_geometry/coherence_bar.png

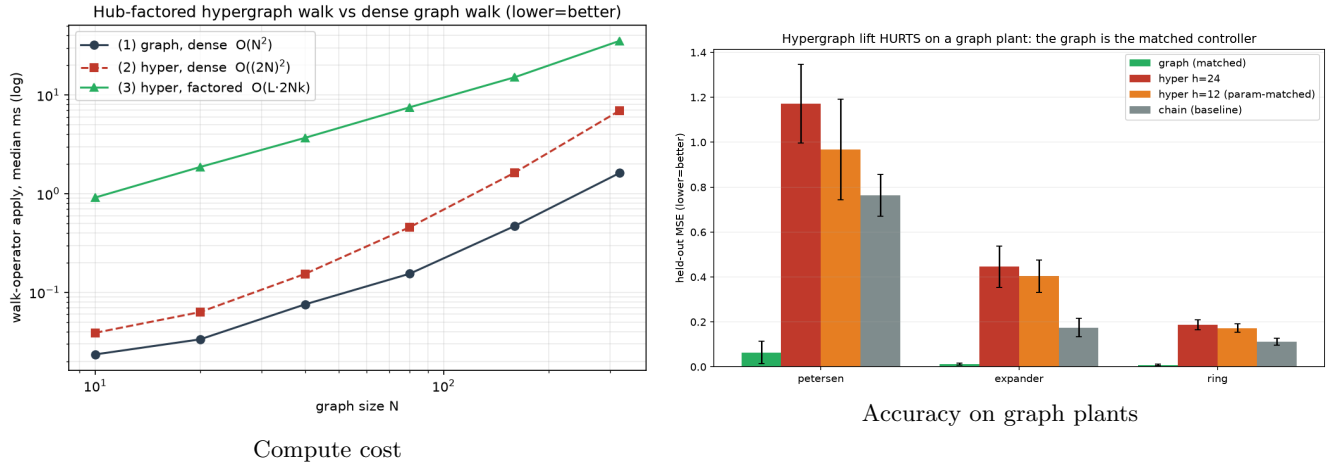


Figure 8: The closed-neighbourhood hypergraph lift costs more and hurts graph plants. The lift is a structural change, not a universal upgrade.

Source: reports/figures/topology_zoo/compute_arity.png;reports/figures/topology_zoo/accuracy_arity.png

Fast RL arm	Reward / accuracy signal	Deploy ms	Params
MLP	-0.157	0.12	3847
SA-HSiKAN	-0.158	0.91	775
HSiKAN-Chebyshev	-0.125	3.63	8071
HSiKAN-CR	-0.133	4.67	8071
Mixture	-0.146	5.17	11833

Table 7: Fast bandit sanity results. SA-HSiKAN is far smaller/faster than vanilla HSiKAN; HSiKAN-Chebyshev can improve reward but costs more rollout time.

Source: reports/2026-06-29-fast-rl-sanity-suite.md

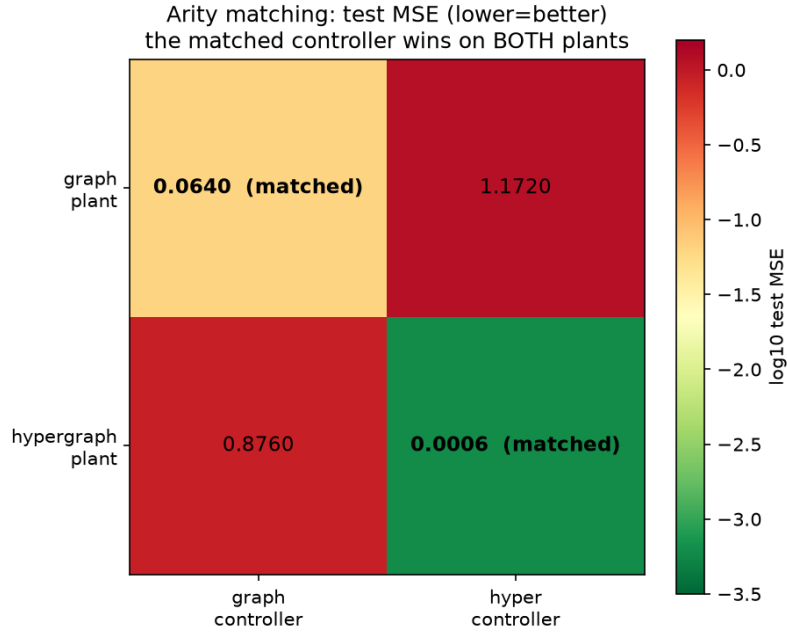


Figure 9: Arity matching. Graph controllers win on graph plants; hypergraph controllers win on hypergraph plants. This is one of the cleanest control-side principles in the repo.

Source: `reports/figures/topology_zoo/arity_matching.png`; `reports/2026-06-29-arity-study.tex`

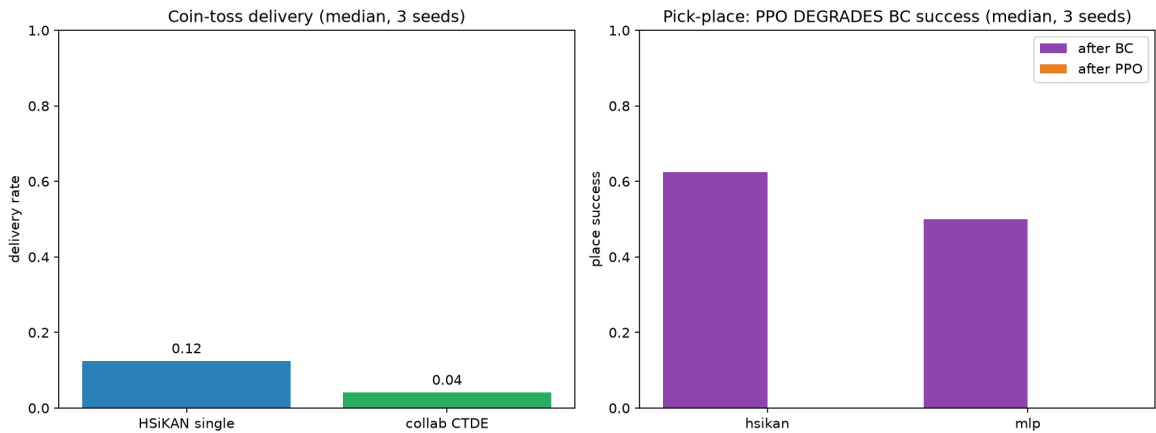


Figure 10: Overnight RL result. Behavior cloning produces useful pick-and-place success; PPO refinement improves scalar return while actual lift/place success collapses. This is reward misalignment, not yet an architecture win.

Source: `reports/figures/overnight_rl.png`; `reports/2026-06-29-rl-overnight.md`

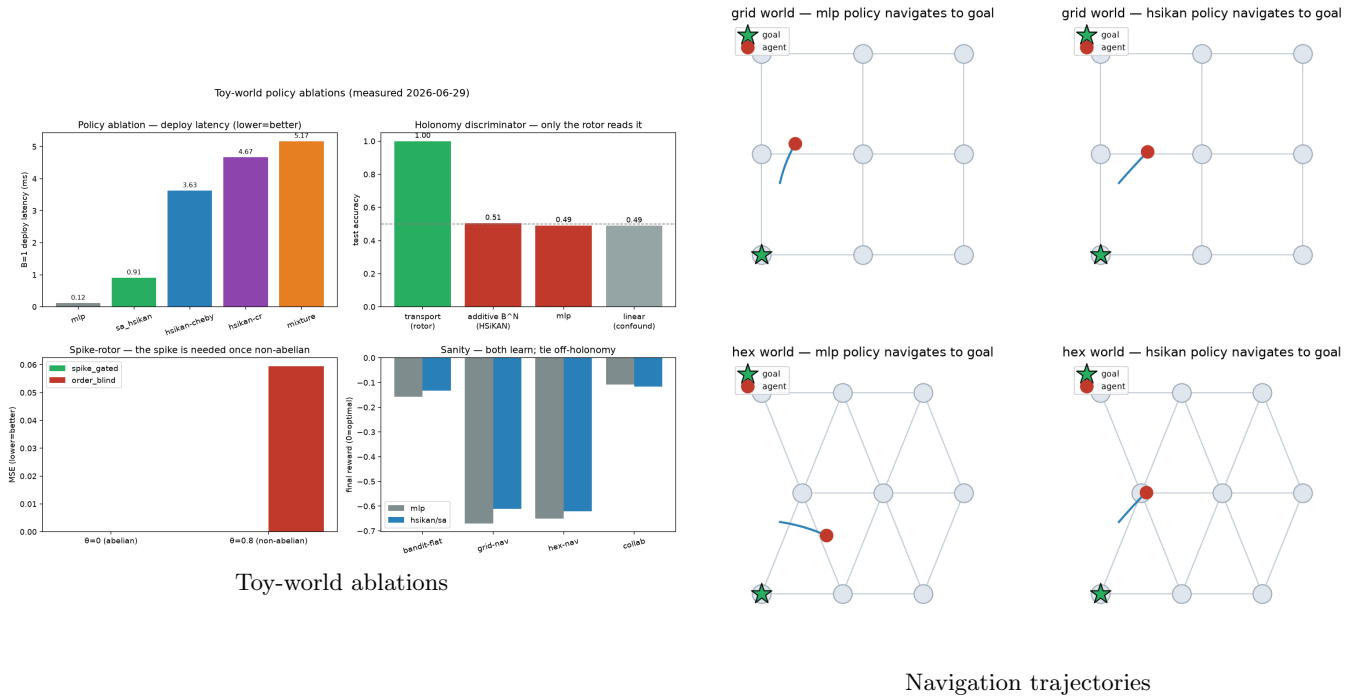


Figure 11: Fast RL sanity worlds. These physics-free tests run real backbones in seconds and are the right front-end before hour-scale MuJoCo runs.

Source: reports/figures/toy_worlds/ablations.png;reports/figures/toy_worlds/nav_trajectories.png

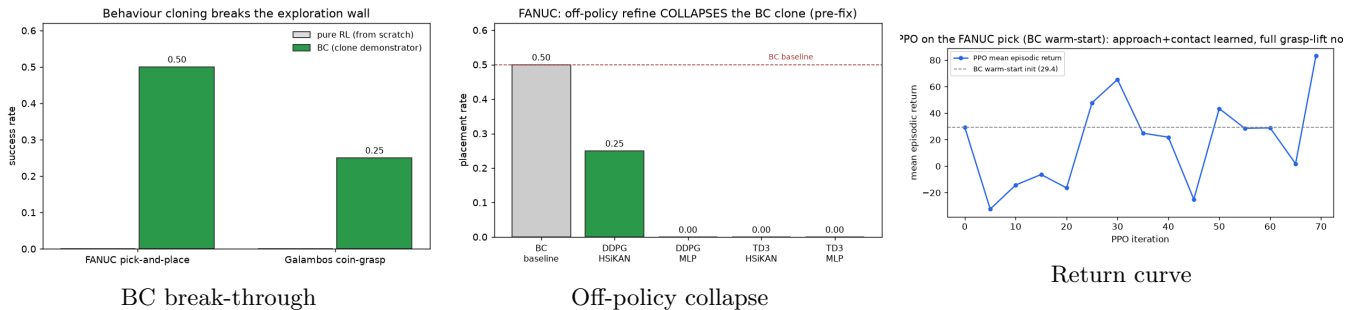


Figure 12: Manipulation learning diagnostics. The visual pattern is consistent: imitation helps; naive refinement can move the scalar objective while damaging task success.

Source: reports/figures/bc_breaks_wall.png;reports/figures/offpolicy_collapse.png;reports/figures/fanuc_ppo_return.png

8 Vision/Soma: readout is the lever

The vision line is best read as a diagnosis story. Early global-mean-pool Soma/RicciStim looked falsified against simple controls. The later readout sweeps show that the structure was often being compressed through the wrong interface: discarding position erased the useful signal. Spatial-pyramid pooling is the current reusable component because it keeps multi-scale position without requiring a fixed flatten head.

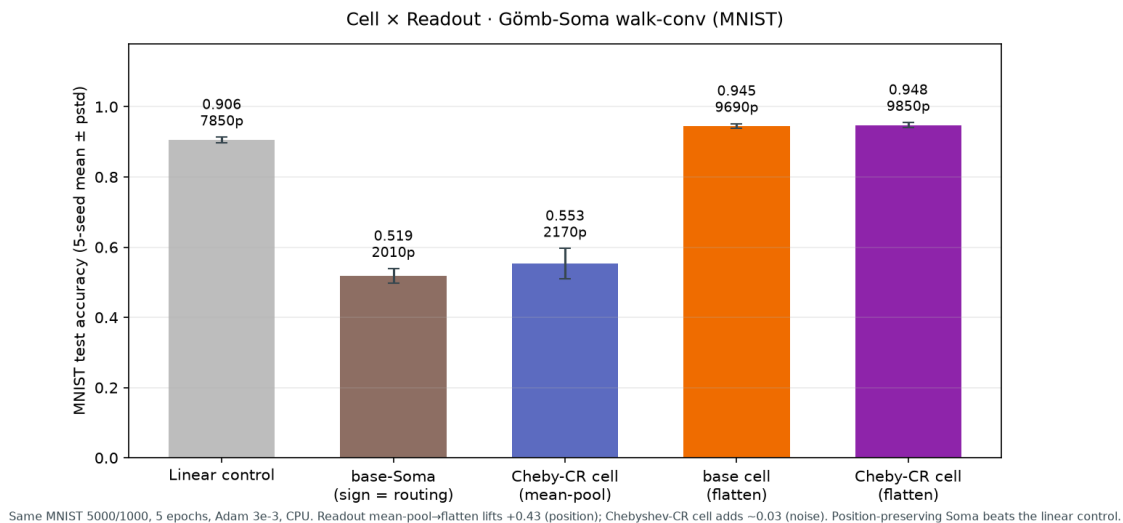


Figure 13: Cell-by-readout sweep. The readout effect is much larger than the cell effect: preserving position can flip Soma from apparent failure to useful signal.

Source: `reports/figures/soma_cheby_cell_readout_20260629.png;reports/2026-06-29-soma-cheby-cell-readout-sweep.md`

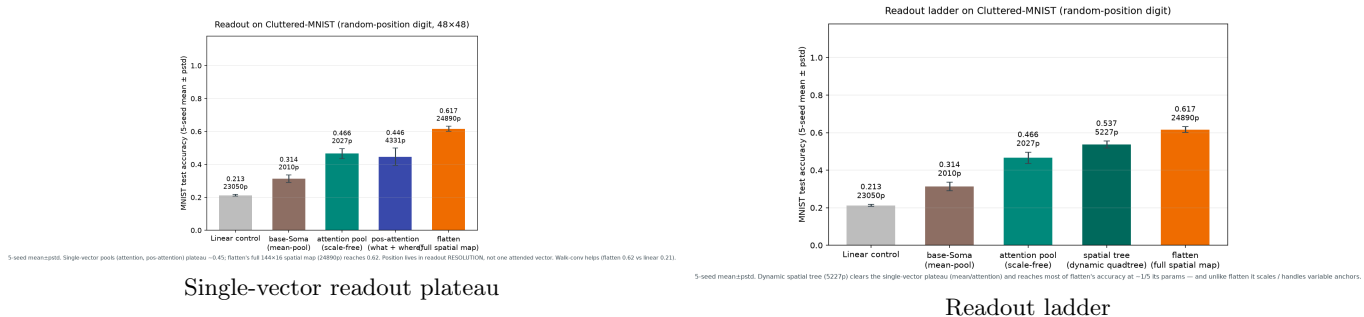


Figure 14: Position-aware readout program. Mean pooling discards location; attention helps; spatial pyramid/tree is the scalable middle; flatten is the fixed-grid upper reference.

Source: `reports/figures/soma_posreadout_cluttered_20260629.png;reports/figures/soma_readout_ladder_20260629.png`

Readout	Parameters	Cluttered-MNIST accuracy
Linear control	23050	0.213
Mean-pool	2010	0.314
Attention	2027	0.466
Spatial tree	5227	0.537
Flatten	24890	0.617

Table 8: Readout ladder as a table. The spatial-tree readout is the scalable knee: about 87 percent of the attention-to-flatten gap at about one fifth of flatten's parameters.

Source: `reports/2026-06-29-soma-spatial-tree-readout.md`



Figure 15: The visual verdict: position-aware spatial pooling helps RicciStim; Z2/U(1) holonomy variants do not become the vision lever.

Source: reports/figures/soma_ricci_tree_readout_20260630.png;reports/figures/soma_holonomy_group_ablation_20260629.png

9 FSR-LM: architecture, quality, and speed

The FSR language-model line is the newest architecture bet. It is not a mature scaling result; it is a tiny, controlled proof-of-concept that separates which parts of the idea currently work. The pre-norm FSR mixer plus HSiKAN channel mix beats a matched transformer on byte-level TinyShakespeare, while the spherical Gomb residual loses. Sparse top-k fixes the long-context memory wall, but fused attention still wins absolute throughput.

Gömb / HSiKAN / Fiber-Spike-Rotor LM (as built, $d = 3n_b$)

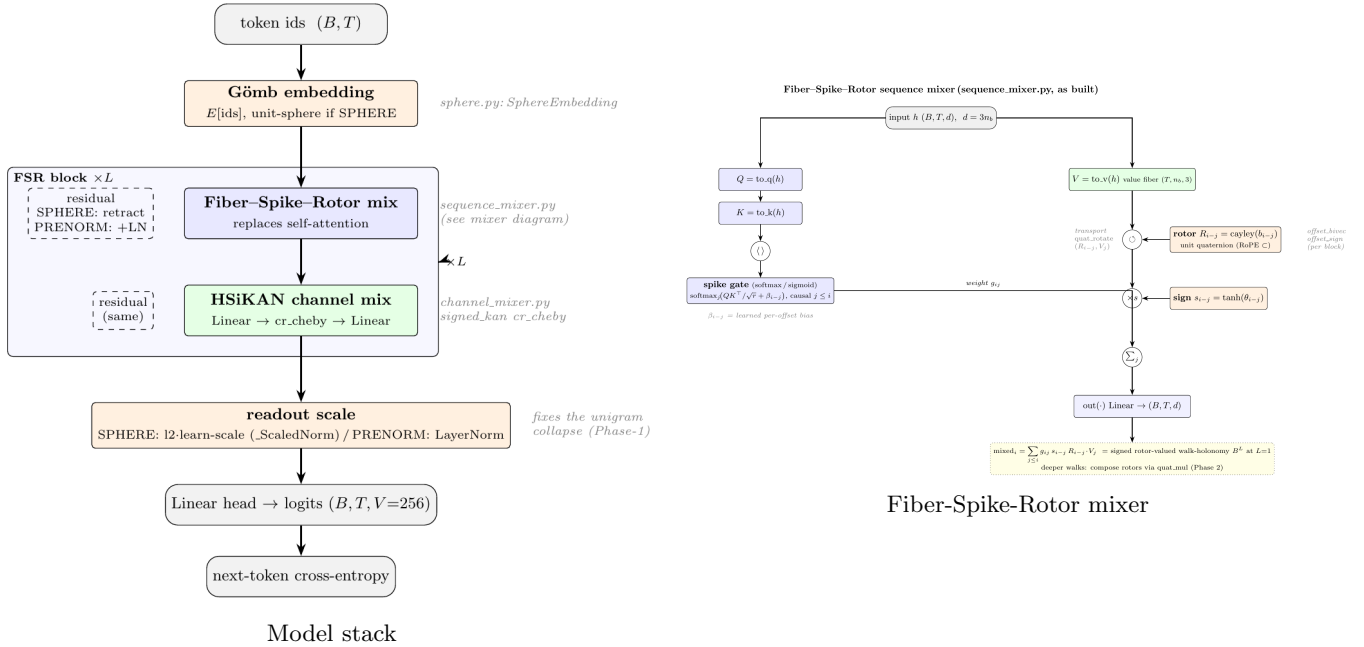


Figure 16: FSR-LM architecture. The sequence mixer is signed, rotor-transported, and spike-gated; the strongest reported variant uses standard pre-norm rather than the Gomb spherical residual.

Source: reports/2026-06-29-fsr-lm-arch-model.png;reports/2026-06-29-fsr-lm-arch-mixer.png

Model	Val bpb median	Tokens/s median	Params
FSR sphere	2.7702	11780	367328
FSR pre-norm	2.4791	11161	368960
Transformer	2.5698	35080	369324

Table 9: FSR Phase 1 result table. The win belongs to FSR pre-norm; the sphere residual loses at full budget.

Source: reports/2026-06-29-fsr-lm-phase1-ab.json

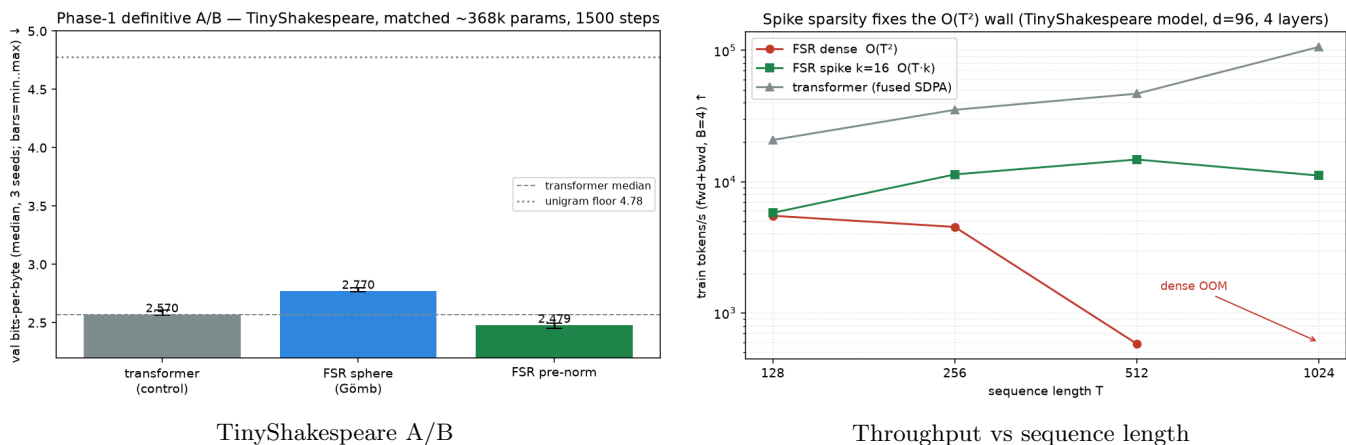


Figure 17: FSR quality and speed. FSR pre-norm beats the matched transformer on tiny byte-LM bpb; hard top-k sparsity prevents the dense long-context wall, but absolute speed still trails fused transformer kernels.

Source: reports/2026-06-29-fsr-lm-phase1.png;reports/2026-06-29-fsr-lm-speed.png

Sequence length	Dense FSR tok/s	FSR $k=16$ tok/s	Transformer tok/s
128	5510	5795	20789
256	4527	11365	35275
512	583	14765	46808
1024	OOM	11138	106329

Table 10: Top-k sparse FSR changes the scaling regime: dense collapses/OOMs while $k=16$ remains feasible.

Source: reports/2026-06-29-fsr-lm-throughput-vs-T.json

10 Holonomy: multiplicative transport is the signal

This section is deliberately small but important. It decides what kind of “structure” the gauge line needs. A cycle-parity label is not an additive aggregate; it is a product accumulated around a loop. The current probe shows that a multiplicative transport reader solves the task exactly, while an additive signed-walk reader, a flat MLP, and a linear guard stay near chance. That supports rotor/transport-style mechanisms for holonomy-heavy tasks and warns against overclaiming plain additive HSiKAN on pure parity.

Arm	Accuracy	Params	Meaning
Transport / rotor-like reader	1.0000	49	Reads the product around the cycle exactly.
Additive signed-walk reader	0.4805	225	The additive HSiKAN-like mechanism does not expose pure parity holonomy here.
MLP	0.5913	1505	Near chance; parity is hard from raw local signs.
Linear guard	0.5054	13	Confirms no local linear leakage.

Table 11: Current holonomy probe run performed during the exploration. This is a decisive visual table: pure cycle holonomy is multiplicative transport, not a plain additive signed-walk sum.

Source: Command: `.\venv\Scripts\python.exe-mhymeko_rl.holonomy_probe--ring-size12--seeds2`

11 Signed-link and rotor line

The signed-link line contains some of the strongest-looking numbers in the repo, but it also contains the strongest warnings about leakage and attribution. The safe framing is: strict protocol design and structural features are valuable; rotor embeddings are parameter-efficient and inductive; the rotor geometry itself is not yet the measured source of AUROC gains. This section therefore prioritizes audit discipline over headline SOTA language.

Question	What the probe answers	What it does not yet answer
Is parity locally visible?	No. Linear guard stays at chance, so the label is not leaking through a local linear cue.	It does not test every possible feature-engineered baseline.
Does additive message passing read pure holonomy?	No. The additive reader is near chance on this task.	It does not falsify HSiKAN on non-parity tasks.
Does transport read holonomy?	Yes. Multiplicative transport reaches 1.0 accuracy in the current run.	Continuous Spin holonomy and aromaticity variants still need full execution.

Table 12: Interpretation guard for the holonomy probe. The result is sharp, but its scope is parity-like cycle holonomy, not all structural learning.

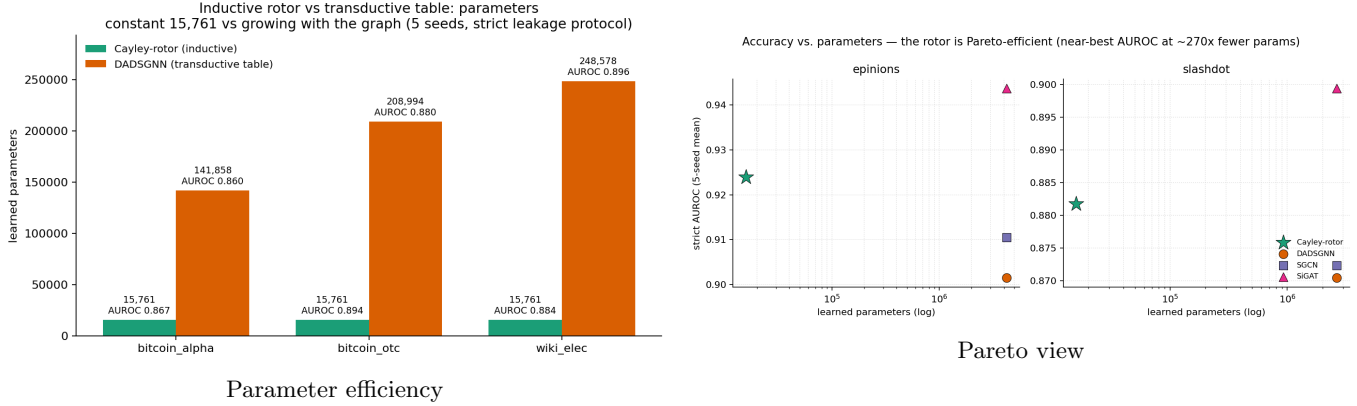


Figure 18: Cayley-rotor signed-link evidence. The honest reading from the reports: rotor embeddings are parameter-efficient and inductive, but later ablations say the rotor geometry itself is not the sole load-bearing ingredient for AUROC. The value is structural features plus strict leakage discipline.

Source: docs/seminar/figures/rotor_param_efficiency.png; docs/seminar/figures/rotor_pareto.png; ROADMAP.md; DONE.md

Model/result family	Alpha	OTC	Slashdot/Epinions	Interpretation
Gomb strict tuned	0.897	0.915	Slashdot 0.902 / Epinions 0.943	Honest strict baseline; avoids the sigma-leakage that inflated earlier cycle numbers.
Optuna-best HSiKAN	0.996	0.993	–	Strong iso-protocol numbers, but reported as transductive/leakage-sensitive and must be framed carefully.
Rotor-vs-MLP embed	tied	tied	tied	Foundational ablation: rotor geometry is not load-bearing on AUROC; structural features are the safer claim.

Table 13: Signed-link takeaway table. The graph-learning line is valuable, but the publishable story is strict protocols and structural priors, not unqualified SOTA claims.

Source: docs/SOTA_RESULTS.md; reports/2026-06-18-rotor-vs-mlp-embed-ablation.md

Risk	Why it matters	Current mitigation
Sigma-product leakage	Cycle products can silently include test-edge sign information under transductive conventions.	Shuffle gates and strict train-edge enumeration identify inflated numbers.
Geometry overclaim	Rotor representations can look novel even when an MLP embedding ties their AUROC.	Rotor-vs-MLP ablation reframes the win as structural features / inductive compactness.
Protocol mismatch	Published baselines may use different dedup or reachability rules.	Protocol-matched SiGAT comparison and planned tier-2/3 audits.

Table 14: Why the signed-link section should be read cautiously. The contribution is stronger when it leads with protocol honesty and parameter efficiency.

12 Policy-as-hypergraph storage

A useful bridge between the framework and RL is that trained policies can become HyMeKo artifacts themselves. The point is not only storing weights compactly; it is making policy provenance, reconstruction, and rendering part of the same hypergraph workflow. This is currently proven on cart-pole-scale policy storage and should be extended to manipulation policies once BC policies are selected as the reliable demo baseline.

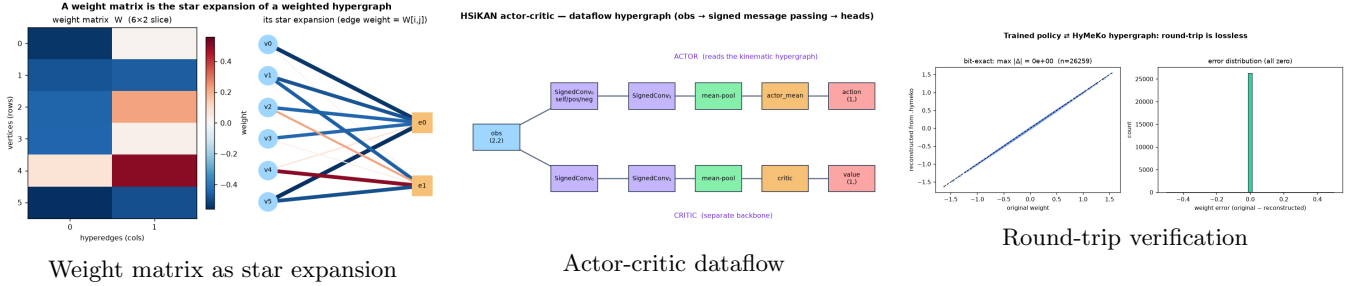


Figure 19: Policy storage as HyMeKo hypergraph. A trained policy can be serialized as a valid HyMeKo graph, reconstructed, and checked bit-exact in the reported cart-pole storage work. This is a bridge between framework and RL: models become first-class graph artifacts.

Source: docs/figures/2026-06-21-policy-storage/;reports/2026-06-21-policy-weight-storage.md

Storage tier	HyMeKo size	External blob	Meaning
T0 decimal	332 KB	none	Human-readable but large.
T1 base64/auto	142–148 KB	none	More compact textual embedding.
T2 content-addressed	9 KB	NPZ blob	Small HyMeKo manifest plus verified external tensor payload.

Table 15: Policy-storage tiers reported on 2026-06-21. The important idea is not compression alone; it is provenance and structural round-trip as part of the model artifact.

Source: DONE.md, 2026-06-21policy-storageentries

13 Thirteen pursuit lanes: what is still missing

The atlas is evidence-heavy, but evidence is only useful if it points to the next decisive measurement. This board lists the thirteen lanes that would convert the repository’s current state into a set of finished claims. Each lane has an existing artifact, a missing discriminator, and a smallest credible next action.

14 Publication and execution matrix

This matrix turns the figures into venue strategy. Different communities should see different leads: formal methods should see cross-view consistency; control should see topology and arity matching; ML architecture should see FSR plus its ablations; robotics collaborators should see BC, scenarios, and demo readiness. Keeping those leads separate prevents one weak line from diluting a stronger one.

15 Visual conclusion

The shortest path through the atlas is not chronological. It is a set of decision figures: one for the publishable framework kernel, one for the structure-matching law, one for the RL warning, one for readout, one for the LM bet, and one for holonomy. These are the figures to use when a collaborator has only a few minutes.

#	Lane	What exists now	Missing discriminator	Near action
1	Cross-view paper	Real CLI emitters, robotics+SysML witnesses, drift demo, proof bundle	Reviewer-proof positioning and final article polish	Submit T-SMC/SMC-style package.
2	CI consistency gate	22-test cross-view suite, extractor strategies	CI integration and optional Gazebo/torch-dataflow extensions	Add verifier as regression gate.
3	Topology matching	8/8 supervised topology map, exact equivariance guard	Closed-loop correlation on real control	Run matched-vs-mismatched controller on cart-pole/arm.
4	Arity matching	Graph-vs-hypergraph plant reversal proves matching law	More plant families and task types	Repeat on contact/gait hypergraph plants.
5	RL refinement	BC works; PPO/off-policy can collapse success	BC-anchor and success-aligned reward proving success does not collapse	Make BC anchor default for refine sweeps.
6	Fast RL lab	Bandit/grid/hex/collab toys; prescreen module	Correlation between toy ranking and MuJoCo ranking	One real-scale confirm per toy-chosen winner.
7	Holonomy	Transport solves parity; additive/MLP near chance	Continuous Spin holonomy and aromaticity versions through production arms	Extend probe beyond T1 parity.
8	StructuralActor	B-power collapse, speed/parameter win, Steiner basis	Closed-loop actuator use, not just supervised target fit	Wire to a control task.
9	FSR-LM	TinyShakespeare bpb win; sparse top-k long-context feasibility	Scale beyond 1 MB corpus, ablations of sign/rotor/spike/cell	Enwik8/text8 subset with spike_k=16.
10	Soma readout	Spatial tree/pyramid improves fixed and variable anchors	Natural-image or cortical-predictivity validation	Export SpatialPyramidPool and test in HyMeYOLO/RL vision.
11	Signed-link	Strict Gomb baseline, leakage audit, rotor Pareto evidence	Final strict baseline table and no-overclaim narrative	Finish tier-2/3 and shuffle audits.
12	Policy storage	Bit-exact policy round-trip and compact T2 storage	Generalize from cart-pole to manipulation policies	Store/render pick-place BC policies as HyMeKo artifacts.
13	Demo narrative	Many figures, GIFs, web views, editor surfaces	One curated path that a collaborator can run end-to-end	Script "one source, many views, one policy" demo.

Table 16: The missing-detail board. These are the thirteen lanes that convert the atlas from evidence inventory into an execution plan.

Scenario / artifact	Current accessible outputs	Main risk visible in atlas	Best next measurement
Pick-and-place	BC/PPO checkpoints, overnight plot, GIF folders	PPO improves return but collapses success	BC-anchor A/B on lift/place success.
Coin-toss / Galambos	Collaborative GIFs, checkpoints, delivery table	CTDE undertrained; single agent currently wins	Longer CTDE refine with anchor.
Planar walker	Walker checkpoint and GIF	Partial forward movement, not full goal reaching	300-iter or shorter-goal confirm.
Cart-pole storage	Policy graph, storage figures, GIFs	Easy task; structure not load-bearing	Use as storage/provenance demo, not architecture proof.
FSR-LM	Phase0/1 plots, architecture diagrams, speed plots	Tiny scale and absolute speed gap	Larger corpus plus fused-kernel decision.
Soma/RicciStim	Readout ladders and Ricci tree figures	Still below flatten and conventional baselines	Spatial pyramid in detection/RL vision.
Cross-view IR	Proof PDF, cross-view figure, JSON verification	Claim must stay about consistency, not raw generator speed	Add CI gate and final paper polish.

Table 17: Scenario matrix: what is runnable or visible now, why it is not yet a finished claim, and the smallest next measurement that would change the state.

Artifact family	Count	Where	Why it matters
Checkpoints	95 files	checkpoints/	The learning lines are not only reports; there are stored policies/models for replay, rendering, and ablation.
Rendered GIFs	91 files	reports/gifs/	The RL/control claims can be inspected behaviorally, not only through scalar metrics.
Plans	100+ dirs	docs/plans/	The project has an explicit plan-first discipline; unfinished work should be judged against these gates.
Reports	345 md	reports/	Most completed tasks have an acceptance note; the atlas samples the strongest, not all.

Table 18: Accessible artifact inventory beyond the embedded figures. The missing work is mostly not lack of code; it is choosing the decisive measurement.

Venue family	Lead claim	Evidence to show visually	Blocking next step
IEEE T-SMC / SMC	Canonical hypergraph IR with machine-verified cross-view consistency	Cross-view grid, systems-engineering table, drift demo, proof bundle	Polish article and supplementary; keep claims centered on consistency.
Formal methods / MDE	Model consistency becomes a machine-checked property, not a convention	Commuting-square diagram, extractor table, Z3 sat/unsat summary	Position against EMF/MLIR/global model management.
Control / CPS	Matched topology and matched arity govern controller representation	Topology heatmap, arity matching plot, StructuralActor speed table	Closed-loop correlation on real control tasks.
RL robotics collaboration	HyMeKo supplies declarative structural RL scenarios and fast pre-screening	Overnight RL figure, toy-world ablations, BC collapse plots	Repair PPO refinement with BC anchor and success-aligned reward.
Graph ML / signed networks	Strict leakage-audited structural features are a methodological contribution	SOTA tables, shuffle gates, transfer grid, rotor-vs-MLP ablation	Finish strict baseline audit and avoid rotor-overclaim.
ML architecture workshop	FSR mixer is a plausible attention alternative at tiny scale	FSR architecture, bpb plot, sparse speed plot	Scale beyond TinyShake-speare; ablate sign/rotor/spike/channel mix.
Vision / neuro	Position-aware readout revives part of Soma/RicciStim, but holonomy is not the vision lever	Readout ladder, Ricci tree transfer, holonomy group ablation	Try stronger scalable readouts and brain-predictivity metrics.

Table 19: Where each evidence block can go, and what is still missing.

If you show one figure	Use the cross-view consistency figure: it is the publishable kernel.
If you show one control result	Use topology map plus arity matching: structure helps when it matches the plant.
If you show one RL warning	Use overnight RL: scalar return improvement can hide success collapse.
If you show one vision result	Use the readout ladder: position-aware pooling is the lever.
If you show one new architecture result	Use FSR bpb plus speed: quality is promising, speed needs fused kernels.
If you show one holonomy result	Use the transport/additive/MLP table: holonomy is multiplicative.

Table 20: The shortest visual route through the entire workspace.